Lecture 7: Unit Roots and Cointegration

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March 27, 2026

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Announcements

- Assignment 2 will be uploaded by the end of today or tomorrow.

  Goal 1: Be able to choose the right framework to use.

  Goal 2: Learn how to implement it in a computer language.

Unit Roots and Cointegration

Introduction

Recap: Asymptotic Theories

- We have studied asymptotic theories for linear regression of time-series data:

$$y_t = x_t'\beta + \varepsilon_t.$$

- The key assumptions are weak stationarity and weak dependence.
- However, most economic data do not exhibit either of these two...
- In practice, we need to verify these two properties before running linear regression.

1. Check if the time-series data are weakly stationary and weakly dependent.
   $\hookrightarrow$ Unit-root test

2-a. If these two properties are satisfied, we can apply linear regression.

2-b. If these two properties are not satisfied, we look into the details.
   $\hookrightarrow$ Cointegration test

Outline

- Preliminary: Statistics
- Unit Roots
- Cointegration

Preliminary: Statistics

Inference

Five-Step Procedure of (One-Sided) Hypothesis Testing

(Step 1): Determine the hypothesis: $H_0 : \theta = \theta_0$ vs. $H_1 : \theta < \theta_0$.

(Step 2): Specify the level of significance α.

(Step 3): Derive the sampling distribution of $\widehat{\theta}$ and construct the test statistic z.

(Step 4): Compute the critical value z_α: $P(|z| < z_\alpha) = 1 - \alpha$.

(Step 5): Reject H_0 if and only if $z < z_\alpha$.

Unit Roots and Cointegration

Unit Roots

Unit Roots

- Three sources of non-weak stationarity (and non-weak dependence):
  1. Time trend
  2. Seasonality
  3. Stochastic trend
     - e.g., a random walk: $X_t = X_0 + \sum_{s=1}^{t} \varepsilon_s$.
- We focus on the stochastic trend.

Unit Roots

- **Insight:** In an AR(1) process,

$$Cov(y_t, y_{t-h}) = \frac{\rho^h}{1-\rho^2}\sigma_\varepsilon^2.$$

- An AR(1) process is weakly stationary

  $\iff |\rho| < 1$ ($\iff |z^*| > 1$)

  $\iff Cov(y_t, y_{t-h}) \to 0$ as $h \to \infty$.

  $\iff$ An AR(1) process is weakly dependent

- Testing for weak stationarity and weak dependence amounts to testing $|z^*| > 1$.

Unit Roots

Definition: Unit Roots

An AR(p) process

$$y_t = \rho_0 + \rho_1 y_{t-1} + \ldots + \rho_p y_{t-p} + \varepsilon_t$$

is said to be a unit root process if its characteristic equation

$$1 - \rho_1 z - \ldots - \rho_p z^p = 0$$

is satisfied with $z^* = 1$.

- A unit root process is not weakly stationary.

  Ex) an AR(1) process ($y_t = \rho_0 + \rho_1 y_{t-1} + \varepsilon_t$) is weakly stationary.
  $\iff 1 - \rho_1 z = 0$ with $|z^*| < 1$.

- An AR(1) process with a unit root = a random walk process.

Testing for a Unit Root: Dickey-Fuller Tests

- Idea: Construct a t-test for an AR(1) process without a drift or a time trend:

$$y_t = \rho y_{t-1} + \varepsilon_t \text{ with } \varepsilon_t \sim IWN(0, \sigma_\varepsilon^2),$$

  for

$$\begin{cases} H_0 : \rho = 1 \\ H_1 : \rho < 1 \end{cases}$$

- Problem: Under H_0, $\{y_t\}$ is not weakly stationary.
  $\Longrightarrow$ We cannot apply the asymptotic theory...
- A way out: Regressing a stationary process onto a non-stationary process.
  - A random walk process is difference stationary.

Testing for a Unit Root: Dickey-Fuller Tests

- The first-difference:

$$\underbrace{y_t - y_{t-1}}_{\Delta y_t} = \alpha + \underbrace{(\rho - 1)}_{=:\theta} y_{t-1} + \varepsilon_t$$

$$\therefore \Delta y_t = \alpha + \theta y_{t-1} + \varepsilon_t$$

- The hypotheses:

$$\begin{cases} H_0 : \rho = 1 \\ H_1 : \rho < 1 \end{cases} \implies \begin{cases} H_0 : \theta = 0 \\ H_1 : \theta < 0 \end{cases}$$

- Under H_0, regressing a stationary process onto a non-stationary process.
  $\implies$ This is manageable, though technically hard...

Testing for a Unit Root: Dickey-Fuller Tests

Proposition: Dickey-Fuller (DF) Test (Dickey, 1976; Fuller, 1976)

The t-statistic for testing

$$H_0 : \rho = 1 \text{ vs. } H_1 : \rho < 1 \text{ for } y_t = \rho y_{t-1} + \varepsilon_t$$
$$\iff H_0 : \theta = 0 \text{ vs. } H_1 : \theta < 0 \text{ for } \Delta y_t = \theta y_{t-1} + \varepsilon_t$$

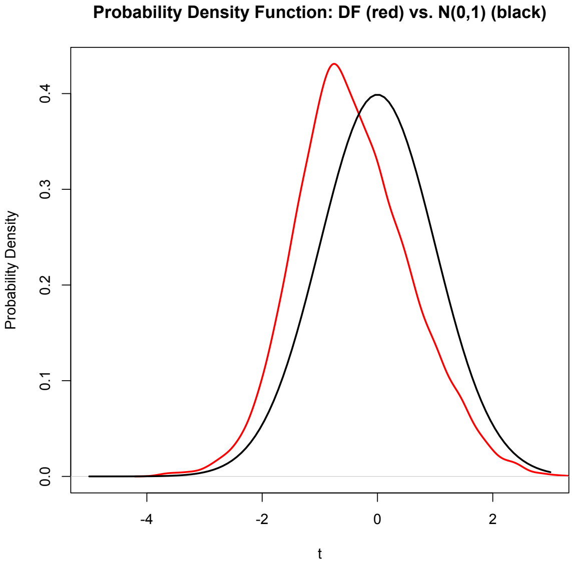
is given by

$$t := \frac{\widehat{\theta} - 0}{se(\widehat{\theta})} = \frac{\widehat{\rho} - 1}{se(\widehat{\rho} - 1)} \xrightarrow{d} DF,$$

where $se(\widehat{\theta}) := (s^2 \sum_{t=1}^{T} y_{t-1}^2)^{1/2}$ with $s^2 := \frac{1}{T-1} \sum_{t=1}^{T} \{y_t - \widehat{\rho} y_{t-1}\}^2$. DF means the Dickey-Fuller distribution.

- The limiting distribution is not a standard normal distribution.
  - It is skewed to the left and has a negative mean.

Testing for a Unit Root: Dickey-Fuller Tests



Testing for a Unit Root: Dickey-Fuller Tests

- Dickey (1976) and Fuller (1996) calculate the critical values for three cases:

(a) An AR(1) process without a drift nor a time trend:

$$y_t = \rho y_{t-1} + \varepsilon_t$$
$$\hookrightarrow \Delta y_t = \theta y_{t-1} + \varepsilon_t$$

(b) An AR(1) process with a drift but without a time trend:

$$y_t = \alpha + \rho y_{t-1} + \varepsilon_t$$
$$\hookrightarrow \Delta y_t = \alpha + \theta y_{t-1} + \varepsilon_t$$

(c) An AR(1) process with a drift and a time trend:

$$y_t = \alpha + \gamma t + \rho y_{t-1} + \varepsilon_t$$
$$\hookrightarrow \Delta y_t = \alpha + \gamma t + \theta y_{t-1} + \varepsilon_t$$

Testing for a Unit Root: Dickey-Fuller Tests

Dickey-Fuller Test Critical Values: (a) An AR(1) process without a drift nor a time trend

Sample size (T)	Probability that the statistic is less than entry							
	0.01	0.025	0.05	0.10	0.90	0.95	0.975	0.99
25	-2.65	-2.26	-1.95	-1.60	0.92	1.33	1.70	2.15
50	-2.62	-2.25	-1.95	-1.61	0.91	1.31	1.66	2.08
100	-2.60	-2.24	-1.95	-1.61	0.90	1.29	1.64	2.04
250	-2.58	-2.24	-1.95	-1.62	0.89	1.28	1.63	2.02
500	-2.58	-2.23	-1.95	-1.62	0.89	1.28	1.62	2.01
∞	-2.58	-2.23	-1.95	-1.62	0.89	1.28	1.62	2.01

Source: Dickey (1976), corrected in Fuller (1996).

Testing for a Unit Root: Dickey-Fuller Tests

Dickey-Fuller Test Critical Values: (b) An AR(1) process with a drift but without a time trend

Sample size (T)	Probability that the statistic is less than entry							
	0.01	0.025	0.05	0.10	0.90	0.95	0.975	0.99
25	-3.75	-3.33	-2.99	-2.64	-0.37	0.00	0.34	0.71
50	-3.59	-3.23	-2.93	-2.60	-0.41	-0.04	0.28	0.66
100	-3.50	-3.17	-2.90	-2.59	-0.42	-0.06	0.26	0.63
250	-3.45	-3.14	-2.88	-2.58	-0.42	-0.07	0.24	0.62
500	-3.44	-3.13	-2.87	-2.57	-0.44	-0.07	0.24	0.61
∞	-3.42	-3.12	-2.86	-2.57	-0.44	-0.08	0.23	0.60

Source: Dickey (1976), corrected in Fuller (1996).

Testing for a Unit Root: Dickey-Fuller Tests

Dickey-Fuller Test Critical Values: (c) An AR(1) process with a drift and a time trend

Sample size (T)	Probability that the statistic is less than entry							
	0.01	0.025	0.05	0.10	0.90	0.95	0.975	0.99
25	-4.38	-3.95	-3.60	-3.24	-1.14	-0.81	-0.50	-0.15
50	-4.16	-3.80	-3.50	-3.18	-1.19	-0.87	-0.58	-0.24
100	-4.05	-3.73	-3.45	-3.15	-1.22	-0.90	-0.62	-0.28
250	-3.98	-3.69	-3.42	-3.13	-1.23	-0.92	-0.64	-0.31
500	-3.97	-3.67	-3.42	-3.13	-1.24	-0.93	-0.65	-0.32
∞	-3.96	-3.67	-3.41	-3.13	-1.25	-0.94	-0.66	-0.32

Source: Dickey (1976), corrected in Fuller (1996).

Testing for a Unit Root: Augmented Dickey-Fuller Tests

- Consider an AR(p) model without a drift or a time trend:

$$y_t = \rho_1 y_{t-1} + \rho_2 y_{t-2} + \ldots + \rho_p y_{t-p} + \varepsilon_t \text{ with } \varepsilon_t \sim IWN(0, \sigma_\varepsilon^2).$$

- This can be transformed to

$$\Delta y_t = \underbrace{(\rho_1 - 1)}_{=:\theta} y_{t-1} + \zeta_1 \Delta y_{t-1} + \zeta_2 \Delta y_{t-2} + \ldots + \zeta_{p-1} \Delta y_{t-(p-1)} + \varepsilon_t.$$

▸ Derivation

- The unit root test boils down to testing

$$H_0 : \rho_1 = 1 \text{ vs. } H_1 : \rho_1 < 1 \qquad \Longleftrightarrow \qquad H_0 : \theta = 0 \text{ vs. } H_1 : \theta < 0$$

- The t statistic converges to the DF distribution:

$$t := \frac{\widehat{\theta} - 0}{se(\widehat{\theta})} = \frac{\widehat{\rho}_1 - 1}{se(\widehat{\rho}_1 - 1)} \xrightarrow{d} DF$$

- This is called the Augmented Dickey-Fuller (ADF) test (Dickey and Fuller, 1979).

Testing for a Unit Root: Augmented Dickey-Fuller Tests

- The critical values vary across the following three cases:

  (a) An AR(p) process without a drift nor a time trend:

$$
\begin{aligned}
&y_t = \rho_1 y_{t-1} + \rho_2 y_{t-2} + \ldots + \rho_p y_{t-p} + \varepsilon_t \\
&\hookrightarrow \Delta y_t = \theta y_{t-1} + \zeta_1 \Delta y_{t-1} + \zeta_2 \Delta y_{t-2} + \ldots + \zeta_{p-1} \Delta y_{t-(p-1)} + \varepsilon_t
\end{aligned}
$$

  (b) An AR(p) process with a drift but without a time trend:

$$
\begin{aligned}
&y_t = \alpha + \rho_1 y_{t-1} + \rho_2 y_{t-2} + \ldots + \rho_p y_{t-p} + \varepsilon_t \\
&\hookrightarrow \Delta y_t = \alpha + \theta y_{t-1} + \zeta_1 \Delta y_{t-1} + \zeta_2 \Delta y_{t-2} + \ldots + \zeta_{p-1} \Delta y_{t-(p-1)} + \varepsilon_t
\end{aligned}
$$

  (c) An AR(p) process with a drift and a time trend:

$$
\begin{aligned}
&y_t = \alpha + \gamma t + \rho_1 y_{t-1} + \rho_2 y_{t-2} + \ldots + \rho_p y_{t-p} + \varepsilon_t \\
&\hookrightarrow \Delta y_t = \alpha + \gamma t + \theta y_{t-1} + \zeta_1 \Delta y_{t-1} + \zeta_2 \Delta y_{t-2} + \ldots + \zeta_{p-1} \Delta y_{t-(p-1)} + \varepsilon_t
\end{aligned}
$$

- The critical values are the same as the DF test.

Testing for a Unit Root: Augmented Dickey-Fuller Tests

- There are no universal rules for the selection of the number of lags, p.
- One option is to use the Akaike Information Criteria:

$$AIC(p) := log\left(\frac{\sum \widehat{\varepsilon}_t^2}{T}\right) + \frac{2(p+1)}{T}.$$

for $y_t = \rho_1 y_{t-1} + \rho_2 y_{t-2} + \ldots + \rho_p y_{t-p} + \varepsilon_t$.

- The optimal lag p^* is given by

$$p^* \in \arg\min_p AIC(p).$$

Testing for a Unit Root: Augmented Dickey-Fuller Tests

- Another is to follow the following sequential t rule:

  (1) Pick a relatively long lag, $\bar{p}$.

  (2) Calculate the t-statistic on lag $\bar{p}$, $t_{\bar{p}}$.

  (3-a) If $t_{\bar{p}}$ is insignificant at some specified critical value,
  $\hookrightarrow$ update $\bar{p} := \bar{p} - 1$ and repeat (2).

  (3-b) If $t_{\bar{p}}$ is significant at some specified critical value,
  $\hookrightarrow$ stop updating and set $p^* := \bar{p}$.

Testing for a Unit Root: Example (Hayashi, 2000)

- The log of U.S. GDP appears to have a linear time trend.

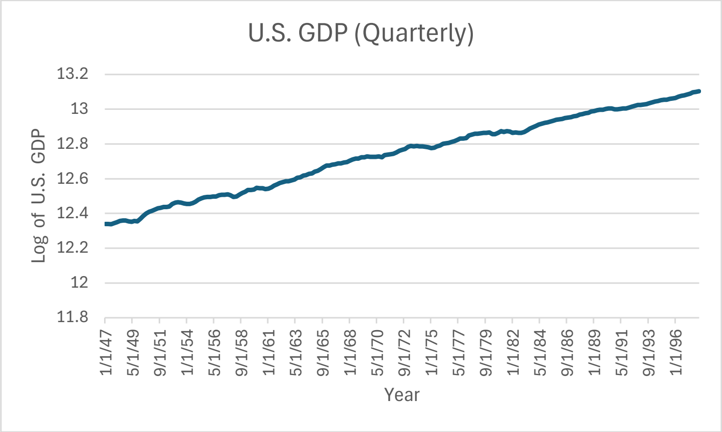


Figure 1: Time-series plot of the U.S. GDP between 1947Q1 and 1998Q1

Testing for a Unit Root: Example (Hayashi, 2000)

- Consider an AR(p) process with a drift and a time trend:

$$
\begin{aligned}
& y_t = \alpha + \gamma t + \rho_1 y_{t-1} + \rho_2 y_{t-2} + \ldots + \rho_p y_{t-p} + \varepsilon_t \\
& \hookrightarrow \Delta y_t = \alpha + \gamma t + \theta y_{t-1} + \zeta_1 \Delta y_{t-1} + \zeta_2 \Delta y_{t-2} + \ldots + \zeta_{p-1} \Delta y_{t-(p-1)} + \varepsilon_t .
\end{aligned}
$$

- The AIC suggests $p^* = 2 \longrightarrow$ effective sample size $= 203$
- The estimated model:

$$
\begin{array}{c}
\Delta y_t = 0.22 + 0.00022\ t - 0.0293\ y_{t-1} + 0.348\ \Delta y_{t-1}. \\
(0.10) \quad (0.00011) \quad (0.0137) \qquad\quad (0.066)
\end{array}
$$

- The ADF t-statistic is

$$
t = \frac{-0.0293 - 0}{0.0137} \approx -2.13
$$

- The 5% critical value: $t_\alpha = -3.45 \sim -3.42$.
- Since $t > t_\alpha$, do not reject H_0 at 5% significance level.

Testing for a Unit Root: Other Tests (Optional)

- Phillips-Perron test (Phillips and Perron, 1988):
  - A version of the ADF test with heteroskedasticity and serial correlations in the error terms
  - The t-statistic follows the same asymptotic distribution as the ADF t-statistic.

- KPSS test (Kwiatkowski et al., 1992):
  - H_0: no unit root vs. H_1: unit root
  - cf) Assignment 1.4

Unit Roots and Cointegration

Cointegration

Overview

- If the unit root test rejects H_0, we can apply the linear regression analysis.
- Q. What if the unit root test does not reject H_0? Can we still use the linear regression?

  A. Yes and No!

- No, because if both $\{y_t\}$ and $\{x_t\}$ are not weakly stationary, the regression coefficients can become statistically significant even if they are independent.
  $\Longrightarrow$ Suprious regression
- Yes, because as long as $\{y_t\}$ and $\{x_t\}$ satisfy a certain relationship, we can still apply the linear regression.
  $\Longrightarrow$ Cointegration

- We need to check cointegration in order to avoid spurious regression.

Suprious Regression

- Granger and Newbold (1974) perform a simulation study.
- Generate two independent random walk processes:

$$x_t = x_{t-1} + u_t \text{ with } u_t \sim GWN(0, \sigma_u^2)$$
$$y_t = y_{t-1} + v_t \text{ with } v_t \sim GWN(0, \sigma_v^2),$$

  where $X_0 = Y_0 = 0$.
- Consider estimating

$$y_t = \beta_0 + \beta_1 x_t + \varepsilon_t$$

  under the standard set of assumptions.
- True value of β_1 is 0.

Suprious Regression

- Hypothesis testing:

$$\begin{cases} H_0 : \beta_1 = 0 \\ H_1 : \beta_1 \neq 0 \end{cases}$$

  at 5% significance level.

- We expect H_0 to be rejected at most 5 times out of 100 times.
- In Granger and Newbold's (1974) study, H_0 is rejected 76 times out of 100 times.
  $\Longrightarrow$ Inference is not reliable...

Suprious Regression

- Q. What is the reason for the spurious regression?
  A. Stochastic trend matters!

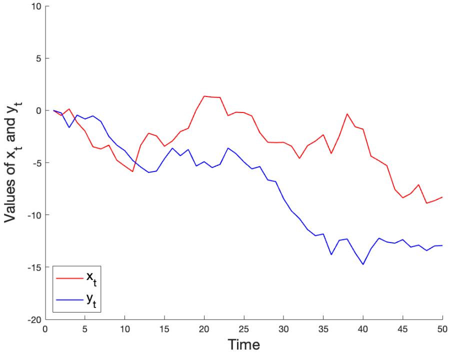


Figure 1: Time-series plots of x_t and y_t

$$x_t = x_{t-1} + u_t = x_0 + \sum_{s=1}^{t} u_s$$

$$y_t = y_{t-1} + v_t = y_0 + \sum_{s=1}^{t} v_s$$

Suprious Regression

- Q. What if there is certainly a linear relationship?
  A. Linearity should hold throughout the entire period! ($\rightarrow$ Cointegration)

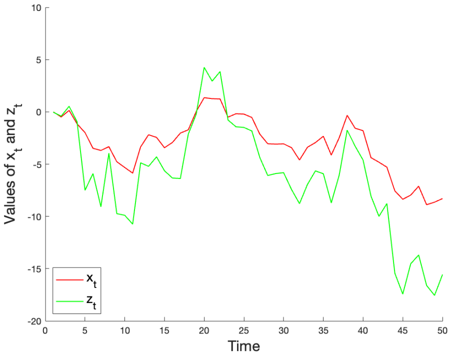


Figure 1: Time-series plots of x_t and z_t

$$x_t = x_{t-1} + u_t$$

$$z_t = \beta_0 + \beta_1 x_t + w_t$$

Cointegration

Definition: Integrated Processes of Order d

(i) A time-series process $\{y_t\}$ is an integrated process of order 0, denoted as $I(0)$, if it is weakly stationary.
(ii) A time-series process $\{y_t\}$ is an integrated process of order d, denoted as $I(d)$, if its dth difference is $I(0)$ (i.e., weakly stationary).

- Ex) A random walk process $\{x_t\}$ is not weakly stationary, but its first-order difference is weakly stationary.
  $\Longrightarrow \{x_t\} \nsim I(0)$ but $\{x_t\} \sim I(1)$.
- In our course, we focus on $I(0)$ and $I(1)$ only.

Cointegration

- The DF and ADF tests can be viewed as testing $I(0)$ vs. $I(1)$:

$$\Delta y_t = \theta y_{t-1} + \varepsilon_t,$$

with

$$\begin{cases} H_0 : \theta = 0 \\ H_1 : \theta < 0 \end{cases} \quad \Longleftrightarrow \quad \begin{cases} H_0 : \{y_t\} \sim I(1) \\ H_1 : \{y_t\} \sim I(0) \end{cases}$$

Cointegration

Definition: Cointegration

Two processes $\{y_t\} \sim I(1)$ and $\{x_t\} \sim I(1)$ are said to be cointegrated if there exists a vector of constants $\gamma \in \mathbb{R}^K$ such that $\{y_t - \gamma' x_t\} \sim I(0)$.

- γ is called a cointegrating vector.
  - This is not necessarily the same as the "regression coefficient" β.
- The existence of γ is often rationalized by the underlying theories:
  - The long-run equilibrium relationship between consumption and investment.
  - The abtrage relationship between financial assets.
- The pair $\{y_t\}$ and $\{x_t\}$ "hang together" over the long run.

Testing for Cointegration

- There are two scenarios depending on whether the cointegrating vector γ is known or not.
- **If γ is known:**

  (Step 1) Define the linear combination with the given values of γ:

$$z_t := y_t - \gamma' x_t.$$

  (Step 2) Apply the DF (or ADF) test for z_t to study $H_0 : z_t \sim I(1)$ vs. $H_1 : z_t \sim I(0)$.

  - If H_0 is rejected, then $\{y_t\}$ and $\{x_t\}$ are cointegrated.

Testing for Cointegration

- **If γ is unknown:**

  (Step 1) Estimate γ by OLS, yeilding $\widehat{\gamma}$.

  (Step 2) Define the linear combination with the estimate $\widehat{\gamma}$:

  $$z_t := y_t - \widehat{\gamma}' x_t.$$

  (Step 3) Apply the DF (or ADF) test for z_t to study $H_0 : z_t \sim I(1)$ vs. $H_1 : z_t \sim I(0)$.

  - If H_0 is rejected, then $\{y_t\}$ and $\{x_t\}$ are cointegrated.

- The OLS estimate $\widehat{\gamma}$ is calculated under H_0 (i.e., a spurious regression).

  $\Longrightarrow$ This issue has to be taken into account in the subsequent DF (or ADF) test.

- Engle and Granger (1987) calculate the critical values for this case.

  - The Engle-Granger test.

Testing for Cointegration

Asymptotic Critical Values for Cointegration Tests

	Probability that the statistic is less than entry			
	0.01	0.025	0.05	0.10
(b) with a drift but without a time trend	-3.90	-3.59	-3.34	-3.04
(c) with a drift and a time trend	-4.32	-4.03	-3.78	-3.50

Source: Engle and Granger (1987) and Phillips and Ouliaris (1990), corrected in Davidson and MacKinnon (1993).

Estimating the Cointegrating Vector

- If (and only if!) the cointegration test rejects H_0, linear regression of y_t onto x_t is meaningful.
- Q. How do we estimate the cointegrating vector?

  A. Two representation theorems:

  1. Dynamic Ordinary Least Squares (DOLS; Stock and Watson, 1993)
     $\hookrightarrow$ long-run relationship
  2. Error Correction Model (ECM; Engle and Granger, 1987)
     $\hookrightarrow$ short-run relationship
- Estimating the cointegrating vector is quite different from testing for cointegration!

Estimating the Cointegrating Vector: DOLS

- If $\{y_t\}$ and $\{x_t\}$ are cointegrated,

$$y_t = \gamma' x_t + \varepsilon_t.$$

  can be transformed to

$$y_t = \gamma' x_t + \sum_{j=-\infty}^{\infty} \delta_j' \Delta x_{t-j} + \varepsilon_t.$$

- In practice, we estimate

$$y_t = \gamma' x_t + \sum_{j=-m}^{m} \delta_j' \Delta x_{t-j} + \varepsilon_t,$$

  where m is selected through the AIC or the sequential t rule.

- The DOLS estimator $\widehat{\gamma}^{DOLS}$ has a set of desirable properties:
  - consistency
  - efficiency
  - asymptotic normal distribution

Estimating the Cointegrating Vector: ECM

- If $\{y_t\}$ and $\{x_t\}$ are cointegrated,

$$y_t = \gamma' x_t + \varepsilon_t$$

can be transformed to

$$\Delta y_t = \delta_0 + \alpha(y_{t-1} - \gamma' x_{t-1}) + \sum_{j=1}^{p} \delta_j' \Delta y_{t-j} + \epsilon_t.$$

- The term $\alpha(y_{t-1} - \gamma' x_{t-1})$ is called the error correction term:

  - $y_{t-1} - \gamma' x_{t-1}$: the deviation from the long-run equilibrium level

  - α: the speed of (short-run) adjustment

Estimating the Cointegrating Vector: ECM

- Usually, $\alpha < 0$:
  - If $y_{t-1} - \gamma' x_{t-1} < 0$,
    $\rightarrow$ y_{t-1} is below the equlibrium level implied by x_{t-1}
    $\rightarrow$ y_t will increase to achieve the equilibrium.
    $\rightarrow$ the error correction term should be positive.
  - If $y_{t-1} - \gamma' x_{t-1} > 0$,
    $\rightarrow$ y_{t-1} is above the equlibrium level implied by x_{t-1}
    $\rightarrow$ y_t will decrease to achieve the equilibrium.
    $\rightarrow$ the error correction term should be negative.

Estimating the Cointegrating Vector: ECM

- Two-step estimation:
  1. Run DOLS to obtain $\widehat{\gamma}^{DOLS}$.
  2. Given $\widehat{\gamma}^{DOLS}$, run the OLS to get $\widehat{\alpha}$:
     - consistency
     - efficiency
     - asymptotic normal distribution

Caveat (Optional)

- Q. What if the cointegration test is not rejected (i.e., no cointegrating relationship)?
- A. There is a generalized notion of cointegration.

Definition: Cointegration of order (d, b)

$\{y_t\}$ and $\{x_t\}$ are cointegrated of order (d, b) if
(i) $\{y_t\} \sim I(d)$ and $\{x_t\} \sim I(d)$; and
(ii) there exists γ such that $\{y_t - \gamma' x_t\} \sim I(d-b)$.

- We have studied a special case where $d = 1$ and $b = 1$.

Summary & Preview

Summary

- We studied two things:
  1. How to check if the data are weakly stationary and weakly dependent.
     $\hookrightarrow$ Unit-root test
  2. What to do if the data turned out not to be weakly stationary and weakly dependent.
     $\hookrightarrow$ Cointegration test

Preview

- The big picture of econometric analysis:
  1. Definition
  2. Data
  3. Identification
  4. Estimation
  5. Inference and Forecasting
- We have focused on identification, estimation, and inference.
- Next, we will study forecasting.

Reference i

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Appendix

Derivation of Augmented Autoregression

- Consider

$$y_t = \rho_0 + \rho_1 y_{t-1} + \rho_2 y_{t-2} + \ldots + \rho_{p-2} y_{t-(p-2)} + \rho_{p-1} y_{t-(p-1)} + \rho_p y_{t-p} + \varepsilon_t .$$

- Add and subtract $\rho_p y_{t-(p-1)}$:

$$\begin{aligned} y_t &= \rho_0 + \rho_1 y_{t-1} + \rho_2 y_{t-2} + \ldots + \rho_{p-2} y_{t-(p-2)} \\ &\quad + \rho_{p-1} y_{t-(p-1)} \underbrace{+ \rho_p y_{t-(p-1)} - \rho_p y_{t-(p-1)}}_{0} + \rho_p y_{t-p} + \varepsilon_t \\ &= \rho_0 + \rho_1 y_{t-1} + \rho_2 y_{t-2} + \ldots + \rho_{p-2} y_{t-(p-2)} \\ &\quad + \underbrace{\rho_{p-1} y_{t-(p-1)} + \rho_p y_{t-(p-1)}}_{(\rho_{p-1}+\rho_p) y_{t-(p-1)}} - \underbrace{\left(\rho_p y_{t-(p-1)} - \rho_p y_{t-p}\right)}_{\rho_p (y_{t-(p-1)} - y_{t-p}) = \rho_p \Delta y_{t-(p-1)}} + \varepsilon_t \\ &= \rho_0 + \rho_1 y_{t-1} + \rho_2 y_{t-2} + \ldots + \rho_{p-2} y_{t-(p-2)} + (\rho_{p-1} + \rho_p) y_{t-(p-1)} - \rho_p \Delta y_{t-(p-1)} + \varepsilon_t . \end{aligned}$$

Derivation of Augmented Autoregression

- Add and subtract $(\rho_{p-1}+\rho_p)y_{t-(p-2)}$:

$$
\begin{aligned}
y_t &= \rho_0 + \rho_1 y_{t-1} + \rho_2 y_{t-2} + \dots \\
&\quad + \rho_{p-2} y_{t-(p-2)} \underbrace{+(\rho_{p-1}+\rho_p)y_{t-(p-2)} - (\rho_{p-1}+\rho_p)y_{t-(p-2)}}_{0} + (\rho_{p-1}+\rho_p)y_{t-(p-1)} \\
&\quad - \rho_p \Delta y_{t-(p-1)} + \varepsilon_t \\
&= \rho_0 + \rho_1 y_{t-1} + \rho_2 y_{t-2} + \dots \\
&\quad + \underbrace{\rho_{p-2} y_{t-(p-2)} + (\rho_{p-1}+\rho_p)y_{t-(p-2)}}_{(\rho_{p-2}+\rho_{p-1}+\rho_p)y_{t-(p-2)}} - \underbrace{\{(\rho_{p-1}+\rho_p)y_{t-(p-2)} - (\rho_{p-1}+\rho_p)y_{t-(p-1)}\}}_{(\rho_{p-1}+\rho_p)(y_{t-(p-2)}-y_{t-(p-1)})=(\rho_{p-1}+\rho_p)\Delta y_{t-(p-2)}} \\
&\quad - \rho_p \Delta y_{t-(p-1)} + \varepsilon_t \\
&= \rho_0 + \rho_1 y_{t-1} + \rho_2 y_{t-2} + \dots \\
&\quad + (\rho_{p-2}+\rho_{p-1}+\rho_p)y_{t-(p-2)} - (\rho_{p-1}+\rho_p)\Delta y_{t-(p-2)} - \rho_p \Delta y_{t-(p-1)} + \varepsilon_t .
\end{aligned}
$$

Derivation of Augmented Autoregression

- Continuing in this fashion,

$$y_t = \rho_0 + \Big(\sum_{i=1}^{p} \rho_i\Big) y_{t-1} + \sum_{i=2}^{p} \beta_i \Delta y_{t-(i-1)} + \varepsilon_t, \text{ where } \zeta_i := -\sum_{j=i}^{p} \rho_j.$$

- Hence,

$$\underbrace{y_t - y_{t-1}}_{\Delta y_t} = \rho_0 + \underbrace{\Big(-1 + \sum_{i=1}^{p} \rho_i\Big)}_{\theta} y_{t-1} + \sum_{i=2}^{p} \zeta_i \Delta y_{t-(i-1)} + \varepsilon_t,$$

  so that

$$\Delta y_t = \rho_0 + \theta y_{t-1} + \sum_{i=2}^{p} \beta_i \Delta y_{t-(i-1)} + \varepsilon_t.$$

- The deterministic term ρ_0 remains untouched through the manipulation.
  $\Longrightarrow$ The same result applies to the cases of $\rho_0 = 0$, $\rho_0 = \alpha$ and $\rho_0 = \alpha + \gamma t$.

Derivation of Augmented Autoregression

- Note that the characteristic equation reads

$$1 - \rho z - \rho_2 z^2 - \ldots - \rho_p z^p = 0.$$

- In particular, under unit root,

$$1 - \rho - \rho_2 - \ldots - \rho_p = 0$$
$$\therefore 1 - \sum_{i=1}^{p} \rho_i = 0.$$

i.e., $\theta = 0$.

back